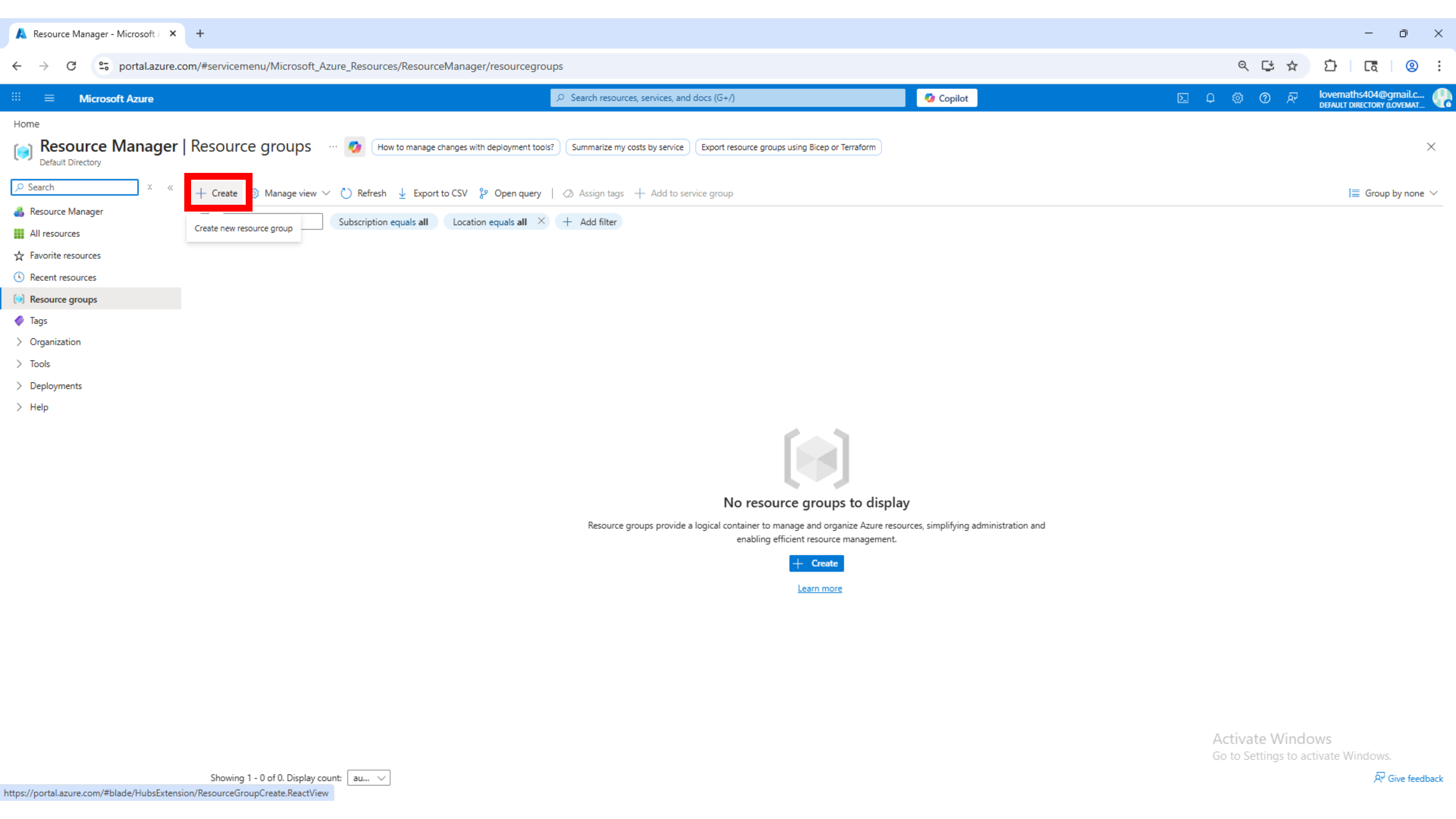
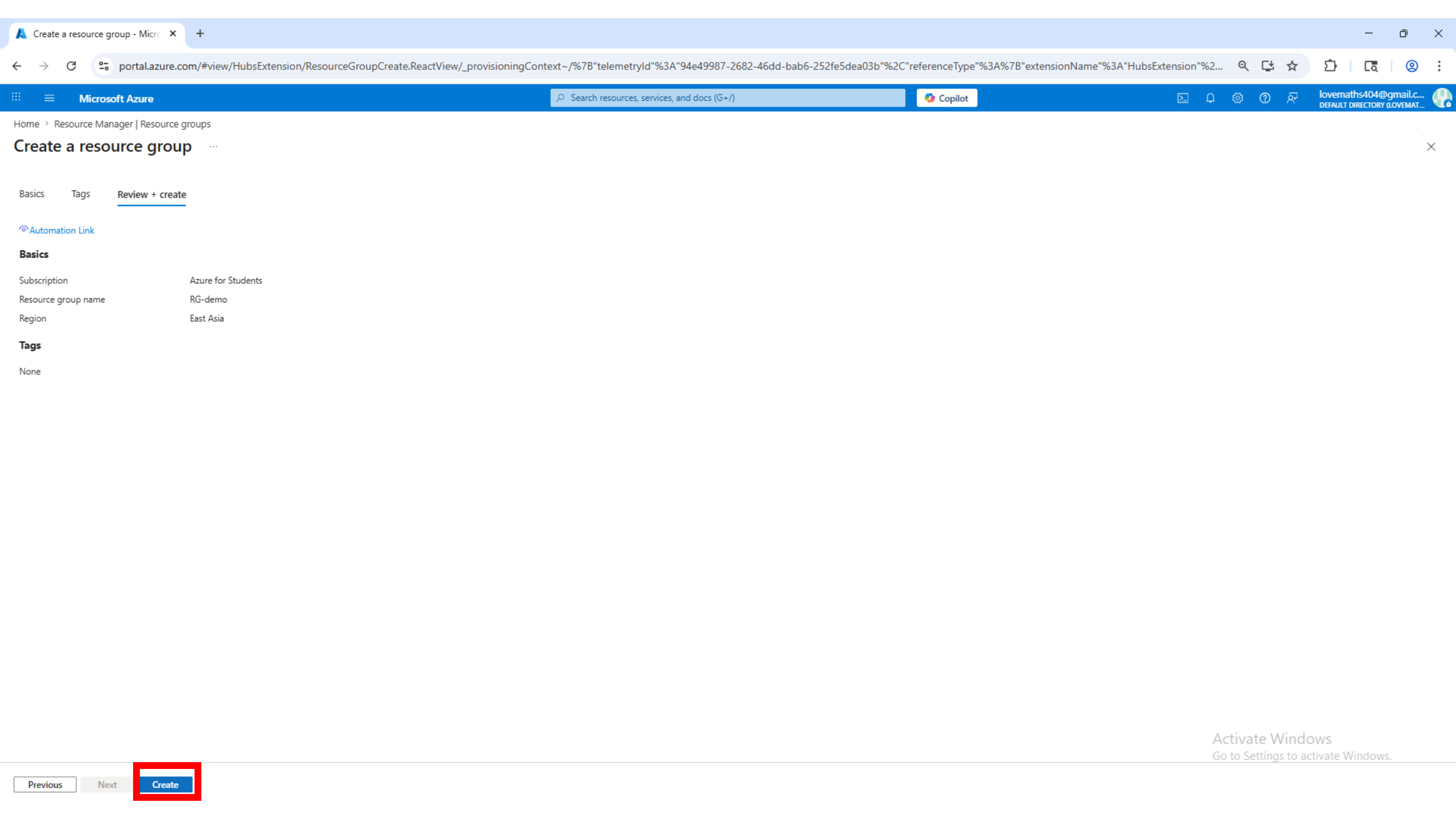


Creating the resource group





Create a resource group ...

Basics Tags Review + create

Automation Link

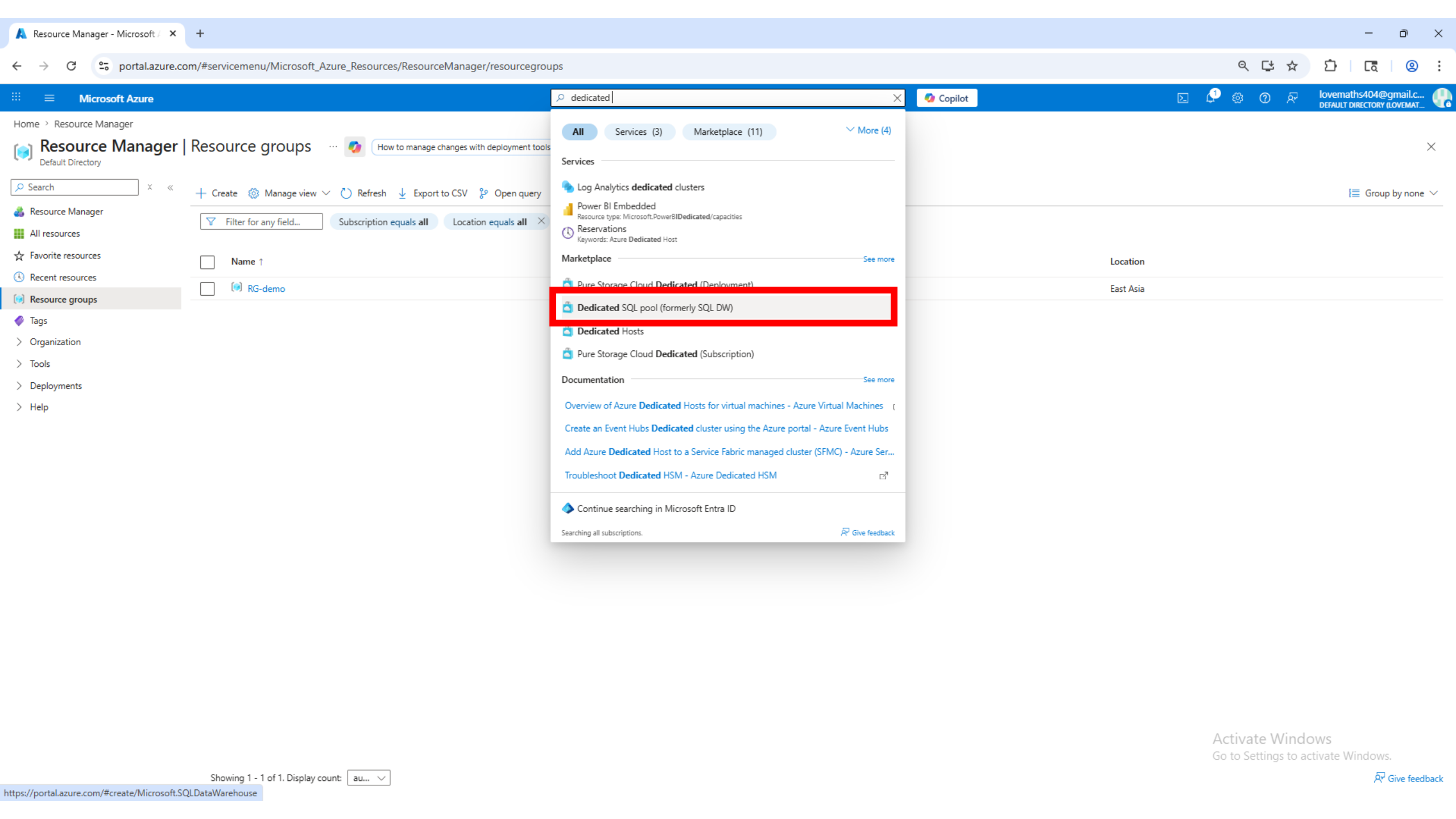
Basics

Subscription	Azure for Students
Resource group name	RG-demo
Region	East Asia

Tags

None

Creating the dedicated sql pool



Create dedicated SQL pool (formerly SQL DW)

Microsoft

Product details

Azure Synapse Analytics
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Est. Cost Per Hour
2.06 USD
[View pricing details](#)

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Azure for Students
Resource group	RG-demo
Region	eastasia
SQL pool name	dedicatedsqlpool
Server	(new) sqlserverdemo786
Storage redundancy	Locally redundant
Authentication method	SQL and Microsoft Entra authentication
Server admin login	sqladmin
Microsoft Entra Admin	lovemaths404@gmail.com
Performance level	Gen2: DW100c

Networking

Allow Azure services and resources to access this server	No
--	----

Additional settings

Use existing data	None
Collation	SQL_Latin1_General_CP1_CI_AS
Advanced data security	Not now

Tags

Microsoft.SQLEDataWarehouse.N

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~-/overview/id/%2Fsubscriptions%2F6796613d-1f98-44b9-a9fe-a1fb80a5dfca%2FresourceGroups%2FRG-demo%2Fproviders%2FMicrosoft.Resources%2Fdeploymen...

Microsoft Azure

Search resources, services, and docs (G+)

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DEFAULT DIRECTORY (LOVEMAT...

Home

Microsoft.SQLEDataWarehouse.NewDatabaseNewServerV4_30004c86ac4348 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

OverviewInputsOutputsTemplate

✔ Your deployment is complete

Deployment name : Microsoft.SQLEDataWarehouse.NewDatabaseNewServerV4_30004c86ac4348

Subscription : Azure for Students

Resource group : RG-demo

Start time : 4/23/2026, 10:05:25 AM

Correlation ID : e9a6493b-e7a3-4c64-b04b-435dece0e0b1

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

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Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

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Adding the client IP
address to allow the
access to sql server

sqlserverdemo786 - Microsoft

portal.azure.com/#@lovemaths404gmail.onmicrosoft.com/resource/subscriptions/6796613d-1f98-44b9-a9fe-a1fb80a5dfca/resourceGroups/RG-demo/providers/Microsoft.Sql/servers/sqlserverdemo786/networking

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Home > Resource Manager | Resource groups > RG-demo > sqlserverdemo786

sqlserverdemo786 | Networking

SQL server

Search

Tags

Quick start

Diagnose and solve problems

Resource visualizer

Settings

Microsoft Entra ID

SQL databases

SQL elastic pools

Properties

Locks

Data management

Backups

Deleted databases

Failover groups

Delete protection (preview)

Import/Export history

Security

Networking

Microsoft Defender for Cloud

Transparent data encryption

Identity

Auditing

Intelligent performance

Monitoring

Automation

Help

Feedback

Public access

Private access

Connectivity

Public network access

Public Endpoints allow access to this resource through the internet using a public IP address. An application or resource that is granted access with the following network rules still requires proper authorization to access this resource. [Learn more](#)

Public network access

☐ Disable

☒ Selected networks

Connections from the IP addresses configured in the Firewall rules section below will have access to this database. By default, no public IP addresses are allowed. [Learn more](#)

Virtual networks

Allow virtual networks to connect to your resource using service endpoints. [Learn more](#)

+ Add a virtual network rule

Rule	Virtual network	Subnet	Address range	Endpoint status	Resource group	Subscription	State
------	-----------------	--------	---------------	-----------------	----------------	--------------	-------

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (49.200.158.22) - Add a firewall rule

Rule name	Start IPv4 address	End IPv4 address
-----------	--------------------	------------------

Exceptions

☐ Allow Azure services and resources to access this server ⓘ

Save

Discard

Activate Windows
Go to Settings to activate Windows.

Home > Resource Manager | Resource groups > RG-demo > sqlserverdemo786

sqlserverdemo786 | Networking

- Search
- Tags
- Quick start
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Microsoft Entra ID
 - SQL databases
 - SQL elastic pools
 - Properties
 - Locks
- Data management
 - Backups
 - Deleted databases
 - Failover groups
 - Delete protection (preview)
 - Import/Export history
- Security
 - Networking
 - Microsoft Defender for Cloud
 - Transparent data encryption
 - Identity
 - Auditing
- Intelligent performance
- Monitoring
- Automation
- Help
- Add or remove favorites by pressing Ctrl+Shift+F

Feedback

Public access Private access Connectivity

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- ☒ Selected networks
- ☐ Connections from the IP addresses configured in the Firewall rules section below will have access to this database. By default, no public IP addresses are allowed. [Learn more](#)

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------	-----------------	--------	---------------	-----------------	----------------	--------------	-------

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (49.200.158.22) + Add a firewall rule

Rule name	Start IPv4 address	End IPv4 address
ClientIPAddress_2026-4-23_10-18-29	49.200.158.22	49.200.158.22

Exceptions

☐ Allow Azure services and resources to access this server

Save Discard

Successfully updated server firewall rules
Successfully updated server firewall rules for server sqlserverdemo786

Activate Windows
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Connecting to the db
server through SSMS

Home > Resource Manager | Resource groups > RG-demo

sqlserverdemo786

SQL server

What is my database's resource usage? Analyze workload increase trends List database permissions

Search Create database New elastic pool New dedicated SQL pool (formerly SQL DW) Import database Reset password Move Delete Feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Quick start
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Microsoft Entra ID
 - SQL databases
 - SQL elastic pools
 - Properties
 - Locks
- Data management
 - Backups
 - Deleted databases
 - Failover groups
 - Delete protection (preview)
 - Import/Export history
- Security
 - Networking
 - Microsoft Defender for Cloud
 - Transparent data encryption
 - Identity
 - Auditing

Essentials

Resource group (move) : [RG-demo](#)
Status : Available
Location : East Asia
Subscription (move) : [Azure for Students](#)
Subscription ID : 6796613d-1f98-44b9-a9fe-a1fb80a5dfca
Tags (edit) : [Add tags](#)

Server admin : sqladmin
Networking : [Show networking settings](#)
Microsoft Entra admin : [Show Microsoft Entra admin settings](#)
Server name : **sqlserverdemo786.database.windows.net**

Notifications (0) Features (6)

All Security (4) Performance (1) Recovery (1)

Microsoft Entra admin
Allows you to centrally manage identity and access to your Azure SQL databases.
CONFIGURED

Microsoft Defender for SQL
Vulnerability Assessment and Advanced Threat Protection.
NOT CONFIGURED

Automatic tuning
Monitors and tunes your database automatically to optimize performance.
CONFIGURED

Auditing
Track database events and writes them to an audit log in Azure storage.
NOT CONFIGURED

Failover groups
Automatically manages replication, connectivity and failover for a set of databases.
NOT CONFIGURED

Transparent data encryption
Encryption at rest for your databases, backups, and logs.
SERVICE-MANAGED KEY

Available resources

Filter by name			All types		
1 dedicated SQL pool (formerly SQL DW)					
Name	↑↓	Type	↑↓	Status	↑↓
Dedicated SQL pool (formerly SQL DW)					
 dedicatedsqlpool		Dedicated SQL pool (formerly SQL DW)		Online	Gen2: DW100c

Activate Windows
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Creating a storage
account and
uploading the csv file

Home > Storage center | Blob Storage

Create a storage account

- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	RG-demo
Location	East Asia
Storage account name	amanstrgacntdemo
Preferred storage type	Azure Blob Storage or Azure Data Lake Storage Gen 2
Primary workload	
Performance	Standard
Account type	Page blobs
Replication	Geo-redundant storage (GRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled

Azure Files

Require Encryption in Transit for SMB	Enabled
---------------------------------------	---------

Security

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in the Azure portal	Disabled

*** Initializing deployment...

Initializing template deployment to resource group 'RG-demo'.

amanstrgacntdemo - Microsoft

portal.azure.com/#@lovemaths404gmail.onmicrosoft.com/resource/subscriptions/6796613d-1f98-44b9-a9fe-a1fb80a5dfca/resourcegroups/RG-demo/providers/Microsoft.Storage/storageAccounts/amanstrgacntdemo/containersList

Microsoft Azure

Search resources, services, and docs (G+)

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DEFAULT DIRECTORY (LOVEMAT...

Home > amanstrgacntdemo_1776920736112 | Overview > amanstrgacntdemo

amanstrgacntdemo

Storage account

Search

«

+ Add container

↑ Upload

↻ Refresh

🗑 Delete

🔒 Change access level

↶ Restore containers

🔧 Edit columns

Search containers by prefix

Only show active containers

Showing all 1 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	Slogs	4/23/2026, 10:38:38 AM	Private	Available	-

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Partner solutions

Resource visualizer

Data storage

Containers

File shares

Queues

Tables

Security + networking

Networking

Access keys

Shared access signature

Encryption

Microsoft Defender for Cloud

Data management

Storage Actions

Redundancy

Data protection

Blob inventory

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Home > amanstrgacntdemo_1776920736112 | Overview > amanstrgacntdemo | Containers

data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

+ Add Directory ↑ Upload ↻ Refresh | 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns

data

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	📁 sales	4/23/2026, 10:40:48 AM				...

✓ Successfully added directory
Successfully added directory 'sales'.

Activate Windows
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Home > amanstrgacntdemo_1776920736112 | Overview > amanstrgacntdemo | Containers

data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns

data > sales

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	[.]						...
☐	sales.csv	4/23/2026, 10:46:05 AM	Hot (Inferred)	Block blob	176 B	Available	...

Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).

Activate Windows
Go to Settings to activate Windows.

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

New QueryExecute

dedicatedsqlpool

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12.0)

Databases

System Databases

dedicatedsqlpool

Tables

Views

External Resources

Programmability

Storage

Security

Integration Services Catalogs

SQLQuery1 - sqladmin (0)

```
1 CREATE TABLE sales (
2     order_id INT,
3     product_name VARCHAR(100),
4     amount FLOAT,
5     order_date DATE
6 )
7 WITH (
8     DISTRIBUTION = HASH(order_id),
9     CLUSTERED COLUMNSTORE INDEX
10 );
```

100 % No issues found Ln: 10, Ch: 3 SPC CRLF Windows 1252

Messages

Commands completed successfully.

Completion time: 2026-04-23T10:56:22.2724452+05:30

100 % No issues found

Query executed successfully.

sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:02 Row: 1, Col: 1 0 rows

GitHub Copilot Chat

New Thread



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I might make mistakes, so check for accuracy.

Query faster with Chat

Explore your query performance with Chat

Explore your database with Chat

Use # for references, / for commands, and @ for extensions.

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Activate Windows

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```
1 CREATE TABLE sales (
2     order_id INT,
3     product_name VARCHAR(100),
4     amount FLOAT,
5     order_date DATE
6 )
7 WITH (
8     DISTRIBUTION = HASH(order_id),
9     CLUSTERED COLUMNSTORE INDEX
10 );
```

100 % No issues found Ln: 10, Ch: 3 SPC CRLF Windows 1252

Messages
Commands completed successfully.
Completion time: 2026-04-23T10:56:22.2724452+05:30

100 % No issues found Ln: 4, Ch: 1 TABS MIXED UTF-8 with BOM

Query executed successfully. sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:02 Row: 1, Col: 1 0 rows

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Connecting the
SSMS with csv file
present in ADLS via
polybase

The Big Picture

PolyBase essentially makes your **external ADLS file look like a SQL table**. To do that safely, it needs a chain of trust and configuration:

Master Key → Credential → External Data Source → File Format → External Table

Each layer depends on the one before it.

1. Master Key

sql

```
CREATE MASTER KEY ENCRYPTION BY PASSWORD = '...';
```

What it is: A symmetric encryption key stored in your database.

Why it's needed: Your SAS token / Access Key is a sensitive secret. SQL Server cannot store it as plain text in the database. The Master Key **encrypts the credential secret at rest** so it's never stored in readable form inside the database files.

Analogy: It's the **lock on the safe** where your credentials are stored. Without the lock, you can't store anything securely inside.

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12)

Databases

System Databases

master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

Constraints

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Statistics

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External Resources

Programmability

Storage

Partition Schemes

Partition Functions

Security

Integration Services Catalogs

SQLQuery1.s...qladmin (0)

```
1 CREATE TABLE dbo.sales (
2     order_id INT,
3     product_name VARCHAR(100),
4     amount FLOAT,
5     order_date DATE
6 )
7 WITH (
8     DISTRIBUTION = HASH(order_id),
9     CLUSTERED COLUMNSTORE INDEX
10 );
11
12 IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE name = '##MS_DatabaseMasterKey##')
13     CREATE MASTER KEY ENCRYPTION BY PASSWORD = 'YourStrongPassword123!';
14
15 IF EXISTS (SELECT * FROM sys.database_scoped_credentials WHERE name = 'storage_cred')
16     DROP DATABASE SCOPED CREDENTIAL storage_cred;
17
18 CREATE DATABASE SCOPED CREDENTIAL storage_cred
19 WITH
20     IDENTITY = 'SHARED ACCESS SIGNATURE',
21     SECRET = 'sv=2025-11-05&ss=bf&srt=sco&sp=rwdlacyx&se=2026-04-30T14:10:46Z&st=2026-04-23T05:55:46Z&spr=https&sig=HDUnQ20iMkhjBOGR0nL%2BMbum';
22
23 IF EXISTS (SELECT * FROM sys.external_data_sources WHERE name = 'adls_source')
24     DROP EXTERNAL DATA SOURCE adls_source;
25
26 CREATE EXTERNAL DATA SOURCE adls_source
27 WITH (
28     TYPE = HADOOP,
29     LOCATION = 'abfss://data@amanstrgacntdemo.dfs.core.windows.net',
30     CREDENTIAL = storage_cred
31 );
32
33 IF EXISTS (SELECT * FROM sys.external_file_formats WHERE name = 'csv_format')
34     DROP EXTERNAL FILE FORMAT csv_format;
35
36 CREATE EXTERNAL FILE FORMAT csv_format
37 WITH (
38
```

Ln: 14, Ch: 73 (162 chars, 2 lines) SPC CRLF Windows 1252

Results

Messages

	name	principal_id	symmetric_key_id	key_length	key_algorithm	algorithm_desc	create_date	modify_date	key_guid	key_thumbprint
1	##MS_DatabaseMasterKey##	1	101	256	A3	AES_256	2026-04-23 05:38:01.427	2026-04-23 05:38:01.427	98F7B900-B4CB-4DE9-967B-73E8F4ACAD9F	NULL

Query executed successfully.

sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:00 Row: 1, Col: 1 1 rows

GitHub Copilot Chat

New Thread

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Explore your database with Chat

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2. Database Scoped Credential

sql

```
CREATE DATABASE SCOPED CREDENTIAL storage_cred  
WITH IDENTITY = 'SHARED ACCESS SIGNATURE', SECRET = 'sv=...';
```

What it is: The actual authentication information (SAS token) encrypted using the Master Key.

Why it's needed: When PolyBase reaches out to ADLS Gen2, it needs to **prove it has permission** to access the storage. The credential is the username/password equivalent for that handshake.

Analogy: It's the **key inside the safe**. The Master Key locks the safe, and the credential is what's stored inside it — your actual proof of identity to Azure Storage.

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12)

Databases

System Databases

master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

Constraints

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Views

External Resources

Programmability

Storage

Partition Schemes

Partition Functions

Security

Integration Services Catalogs

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11
12
13 IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE name = '##MS_DatabaseMasterKey##')
14     CREATE MASTER KEY ENCRYPTION BY PASSWORD = 'YourStrongPassword123!';
15
16 IF EXISTS (SELECT * FROM sys.database_scoped_credentials WHERE name = 'storage_cred')
17     DROP DATABASE SCOPED CREDENTIAL storage_cred;
18
19 CREATE DATABASE SCOPED CREDENTIAL storage_cred
20 WITH
21     IDENTITY = 'SHARED ACCESS SIGNATURE',
22     SECRET = 'sv=2025-11-05&ss=bf&srt=sco&sp=rwdlacyx&se=2026-04-30T14:10:46Z&st=2026-04-23T05:55:46Z&spr=https&sig=HDUnQ20iMkhjBOGR0nL%2BMbum';
23
24 IF EXISTS (SELECT * FROM sys.external_data_sources WHERE name = 'adls_source')
25     DROP EXTERNAL DATA SOURCE adls_source;
26
27 CREATE EXTERNAL DATA SOURCE adls_source
28 WITH (
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30     LOCATION = 'abfss://data@amanstrgacntdemo.dfs.core.windows.net',
31     CREDENTIAL = storage_cred
32 );
33
34 IF EXISTS (SELECT * FROM sys.external_file_formats WHERE name = 'csv_format')
35     DROP EXTERNAL FILE FORMAT csv_format;
36
37 CREATE EXTERNAL FILE FORMAT csv_format
38 WITH (
```

Ln: 23, Ch: 1 (457 chars, 7 lines) SPC CRLF Windows 1252

Results

Messages

	name	principal_id	symmetric_key_id	key_length	key_algorithm	algorithm_desc	create_date	modify_date	key_guid	key_thumbprint
1	##MS_DatabaseMasterKey##	1	101	256	A3	AES_256	2026-04-23 05:38:01.427	2026-04-23 05:38:01.427	98F7B900-B4CB-4DE9-967B-73E8F4ACAD9F	NULL

Query executed successfully.

sqlserverdemo786.database.w... | sqladmin (0) | dedicatedsqlpool | 00:00:00 | Row: 1, Col: 1 | 1 rows

GitHub Copilot Chat

New Thread

Copilot Chat

I might make mistakes, so check for accuracy.

Query faster with Chat

Explore your query performance with Chat

Explore your database with Chat

Use # for references, / for commands, and @ for extensions.

Ask Copilot

Activate Windows

Go to Settings to activate Windows.

3. External Data Source

sql



```
CREATE EXTERNAL DATA SOURCE adls_source
WITH (
    TYPE = HADOOP,
    LOCATION = 'abfss://data@amanstrgacntdemo.dfs.core.windows.net',
    CREDENTIAL = storage_cred
);
```

What it is: A named pointer to your storage location, linked to the credential.

Why it's needed: It tells PolyBase **where the data lives** and **which credential to use** to get there. Without this, every external table would need the full storage URL and credential repeated every time.

Why TYPE = HADOOP: ADLS Gen2 uses the Hadoop Distributed File System (HDFS) protocol internally. Even though it's Azure storage, PolyBase communicates with it using the Hadoop connector which supports the `abfss://` protocol natively.

Analogy: It's the **address and access pass** combined — like a pre-configured VPN connection to your storage account.

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12)

Databases

System Databases

master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

Constraints

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Security

Integration Services Catalogs

SQLQuery1.s...qladmin (0)*

```
13 IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE name = '##MS_DatabaseMasterKey##')
14     CREATE MASTER KEY ENCRYPTION BY PASSWORD = 'YourStrongPassword123!';
15
16 IF EXISTS (SELECT * FROM sys.database_scoped_credentials WHERE name = 'storage_cred')
17     DROP DATABASE SCOPED CREDENTIAL storage_cred;
18
19 CREATE DATABASE SCOPED CREDENTIAL storage_cred
20 WITH
21     IDENTITY = 'SHARED ACCESS SIGNATURE',
22     SECRET = 'sv=2025-11-05&ss=bf&srt=sco&sp=rwdlacyx&se=2026-04-30T14:10:46Z&st=2026-04-23T05:55:46Z&spr=https&sig=HDUnQ20iMkhjBOGR0nL%2BMbum';
23
24 IF EXISTS (SELECT * FROM sys.external_data_sources WHERE name = 'adls_source')
25     DROP EXTERNAL DATA SOURCE adls_source;
26
27 CREATE EXTERNAL DATA SOURCE adls_source
28 WITH (
29     TYPE = HADOOP,
30     LOCATION = 'abfss://data@amanstrgacntdemo.dfs.core.windows.net',
31     CREDENTIAL = storage_cred
32 );
33
34 IF EXISTS (SELECT * FROM sys.external_file_formats WHERE name = 'csv_format')
35     DROP EXTERNAL FILE FORMAT csv_format;
36
37 CREATE EXTERNAL FILE FORMAT csv_format
38 WITH (
39     FORMAT_TYPE = DELIMITEDTEXT,
40     FORMAT_OPTIONS (
41         FIELD_TERMINATOR = ',',
42         STRING_DELIMITER = '"',
43         FIRST_ROW = 2, -- skips header row
44         USE_TYPE_DEFAULT = FALSE
45     )
46 );
47
48 IF OBJECT_ID('dbo.sales_external') IS NOT NULL
49     DROP EXTERNAL TABLE dbo.sales_external;
50
```

Ln: 32, Ch: 3 (298 chars, 9 lines) SPC CRLF Windows 1252

Results

Messages

	name	principal_id	symmetric_key_id	key_length	key_algorithm	algorithm_desc	create_date	modify_date	key_guid	key_thumbprint
1	##MS_DatabaseMasterKey##	1	101	256	A3	AES_256	2026-04-23 05:38:01.427	2026-04-23 05:38:01.427	98F7B900-B4CB-4DE9-967B-73E8F4ACAD9F	NULL

Query executed successfully.

sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:00 Row: 1, Col: 1 1 rows

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4. External File Format

sql

```
CREATE EXTERNAL FILE FORMAT csv_format
WITH (
    FORMAT_TYPE = DELIMITEDTEXT,
    FORMAT_OPTIONS (
        FIELD_TERMINATOR = ',',
        STRING_DELIMITER = '"',
        FIRST_ROW = 2,
        USE_TYPE_DEFAULT = FALSE
    )
);
```

What it is: A reusable definition that describes how the file is structured.

Why it's needed: PolyBase reads raw bytes from ADLS — it has no idea whether the file is CSV, Parquet, ORC, or JSON. The file format tells it **how to parse and interpret those bytes** into rows and columns.

Why each option matters:

Option	Purpose
FIELD_TERMINATOR = ','	Tells PolyBase columns are separated by comma
STRING_DELIMITER = '"'	Values wrapped in quotes are treated as single field
FIRST_ROW = 2	Skip row 1 (header) and start reading data from row 2
USE_TYPE_DEFAULT = FALSE	Throw error if NULL found instead of silently defaulting



FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

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Connect

sqlserverdemo786.database.windows.net (SQL Server 12)

Databases

System Databases

master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

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SQLQuery1.s...qladmin (0)*

```
22 SECRET = 'sv=2025-11-05&ss=bf&srt=sco&sp=rwdlacyx&se=2026-04-30T14:10:46Z&st=2026-04-23T05:55:46Z&spr=https&sig=HDUnQ20iMkhjBOGR0nL%2BMbum
23
24 IF EXISTS (SELECT * FROM sys.external_data_sources WHERE name = 'adls_source')
25 DROP EXTERNAL DATA SOURCE adls_source;
26
27 CREATE EXTERNAL DATA SOURCE adls_source
28 WITH (
29 TYPE = HADOOP,
30 LOCATION = 'abfss://data@amanstrgacntdemo.dfs.core.windows.net',
31 CREDENTIAL = storage_cred
32 );
33
34 IF EXISTS (SELECT * FROM sys.external_file_formats WHERE name = 'csv_format')
35 DROP EXTERNAL FILE FORMAT csv_format;
36
37 CREATE EXTERNAL FILE FORMAT csv_format
38 WITH (
39 FORMAT_TYPE = DELIMITEDTEXT,
40 FORMAT_OPTIONS (
41 FIELD_TERMINATOR = ',',
42 STRING_DELIMITER = '"',
43 FIRST_ROW = 2, -- skips header row
44 USE_TYPE_DEFAULT = FALSE
45 )
46 );
47
48 IF OBJECT_ID('dbo.sales_external') IS NOT NULL
49 DROP EXTERNAL TABLE dbo.sales_external;
50
51 CREATE EXTERNAL TABLE dbo.sales_external
52 (
53 order_id INT,
54 product_name VARCHAR(100),
55 amount FLOAT,
56 order_date DATE
57 )
58 WITH (
59 LOCATION = 'sales/sales.csv',
```

100 % No issues found Ln: 46, Ch: 3 (388 chars, 13 lines) SPC CRLF Windows 1252

Results

Messages

	name	principal_id	symmetric_key_id	key_length	key_algorithm	algorithm_desc	create_date	modify_date	key_guid	key_thumbprint
1	##MS_DatabaseMasterKey##	1	101	256	A3	AES_256	2026-04-23 05:38:01.427	2026-04-23 05:38:01.427	98F7B900-B4CB-4DE9-967B-73E8F4ACAD9F	NULL

Query executed successfully.

sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:00 Row: 1, Col: 1 1 rows

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5. External Table

sql

```
CREATE EXTERNAL TABLE dbo.sales_external (...)  
WITH (  
    LOCATION = 'sales/sales.csv',  
    DATA_SOURCE = adls_source,  
    FILE_FORMAT = csv_format  
);
```

What it is: A virtual table that maps your CSV file to SQL columns.

Why it's needed: SQL Server only understands tables — it doesn't know how to query a file path directly. The external table acts as a **bridge** that makes the CSV file queryable using standard SQL. It combines:

- The **Data Source** (where to find the file)
- The **File Format** (how to read it)
- The **Column definitions** (how to map file columns to SQL types)

Important: No data is copied into SQL at this point. The external table is just a **view over the file** — data stays in ADLS.

Analogy: It's like a **window into the file** — you're not bringing the file into SQL, you're just making SQL able to see through the window and read it.

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12)

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master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

Constraints

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SQLQuery1.s...qladmin (0)*

```
37 CREATE EXTERNAL FILE FORMAT csv_format
38 WITH (
39     FORMAT_TYPE = DELIMITEDTEXT,
40     FORMAT_OPTIONS (
41         FIELD_TERMINATOR = ',',
42         STRING_DELIMITER = '|',
43         FIRST_ROW = 2, -- skips header row
44         USE_TYPE_DEFAULT = FALSE
45     )
46 );
47
48 IF OBJECT_ID('dbo.sales_external') IS NOT NULL
49     DROP EXTERNAL TABLE dbo.sales_external;
50
51 CREATE EXTERNAL TABLE dbo.sales_external
52 (
53     order_id INT,
54     product_name VARCHAR(100),
55     amount FLOAT,
56     order_date DATE
57 )
58 WITH (
59     LOCATION = 'sales/sales.csv',
60     DATA_SOURCE = adls_source,
61     FILE_FORMAT = csv_format
62 );
63
64 INSERT INTO dbo.sales
65 SELECT
66     order_id,
67     product_name,
68     amount,
69     order_date
70 FROM dbo.sales_external;
71
72 select * from dbo.sales;
73
```

100 %

No issues found

Ln: 63, Ch: 1 (367 chars, 15 lines) SPC CRLF Windows 1252

Results

Messages

	name	principal_id	symmetric_key_id	key_length	key_algorithm	algorithm_desc	create_date	modify_date	key_guid	key_thumbprint
1	##MS_DatabaseMasterKey##	1	101	256	A3	AES_256	2026-04-23 05:38:01.427	2026-04-23 05:38:01.427	98F7B900-B4CB-4DE9-967B-73E8F4ACAD9F	NULL

Query executed successfully.

sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:00 Row: 1, Col: 1 1 rows

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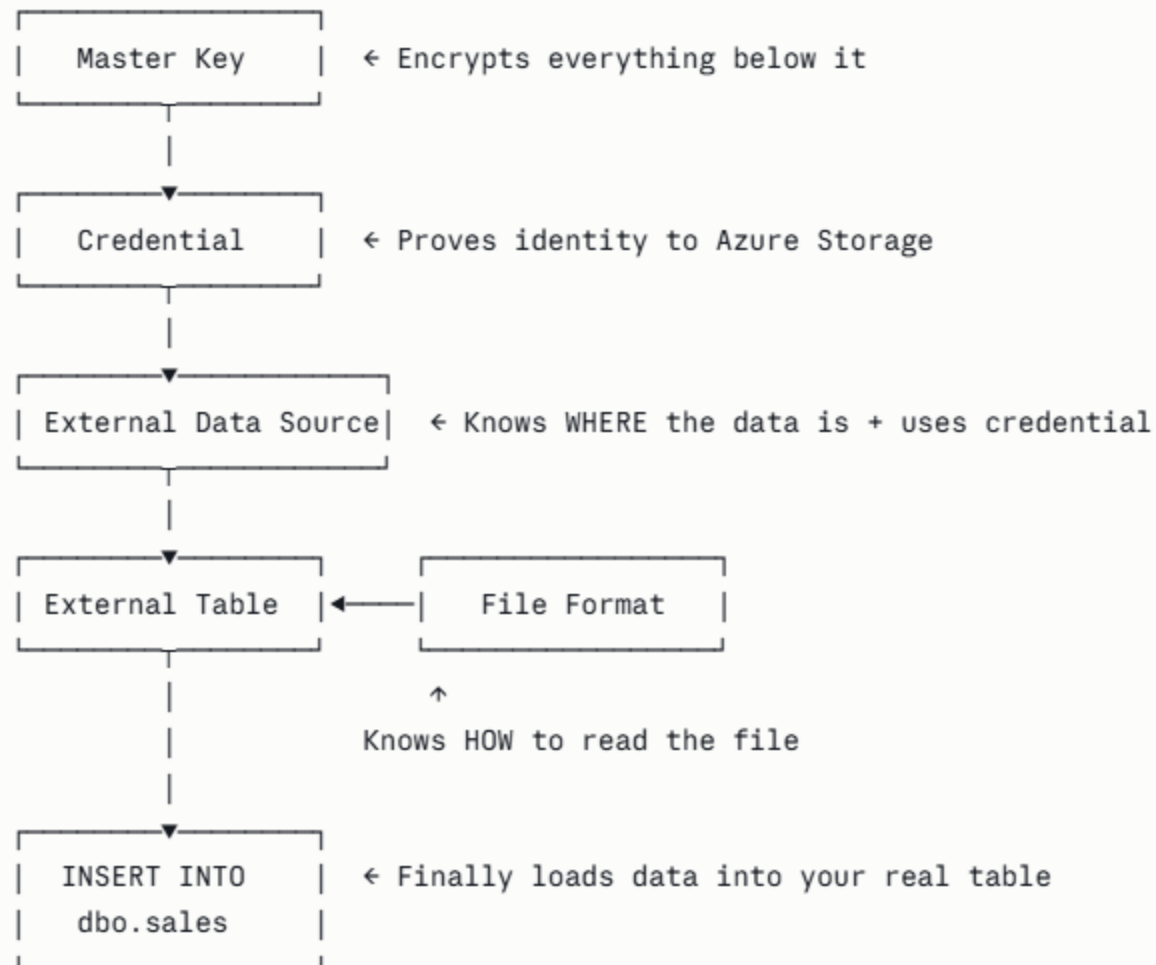
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How They All Chain Together



Each component has a **single responsibility** — security, location, parsing, or mapping — and together they form a complete pipeline to bring external data safely into your SQL pool.

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12.0)

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System Databases

master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

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SQLQuery1.s...qladmin (0)*

55 amount FLOAT,
56 order_date DATE
57)
58 WITH (
59 LOCATION = 'sales/sales.csv',
60 DATA_SOURCE = adls_source,
61 FILE_FORMAT = csv_format
62);
63
64
65 INSERT INTO dbo.sales
66 SELECT
67 order_id,
68 product_name,
69 amount,
70 order_date
71 FROM dbo.sales_external;
72
73 select * from dbo.sales;

100 % No issues found Ln: 71, Ch: 25 (119 chars, 7 lines) SPC CRLF Windows 1252

Results

Messages

	name	principal_id	symmetric_key_id	key_length	key_algorithm	algorithm_desc	create_date	modify_date	key_guid	key_thumbprint
1	##MS_DatabaseMasterKey##	1	101	256	A3	AES_256	2026-04-23 05:38:01.427	2026-04-23 05:38:01.427	98F7B900-B4CB-4DE9-967B-73E8F4ACAD9F	NULL

Query executed successfully. sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:00 Row: 1, Col: 1 1 rows

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Result

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

dedicatedsqlpool

Execute

Object Explorer

Connect

sqlserverdemo786.database.windows.net (SQL Server 12)

Databases

System Databases

master

dedicatedsqlpool

Tables

External Tables

dbo.sales

Columns

order_id (int, null)

product_name (varchar(100), null)

amount (float, null)

order_date (date, null)

Constraints

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SQLQuery1.s...qladmin (0)*

55 amount FLOAT,

56 order_date DATE

57)

58 WITH (

59 LOCATION = 'sales/sales.csv',

60 DATA_SOURCE = adls_source,

61 FILE_FORMAT = csv_format

62);

63

64

65 INSERT INTO dbo.sales

66 SELECT

67 order_id,

68 product_name,

69 amount,

70 order_date

71 FROM dbo.sales_external;

72

73 select * from dbo.sales;

100 %

No issues found

Ln: 73, Ch: 25 (24 chars, 1 lines) SPC CRLF Windows 1252

Results

Messages

	order_id	product_name	amount	order_date
1	1	Laptop	55000	2024-01-10
2	5	Headphones	2500	2024-01-14
3	2	Mouse	500	2024-01-11
4	3	Keyboard	1500	2024-01-12
5	4	Monitor	12000	2024-01-13

Query executed successfully.

sqlserverdemo786.database.w... sqladmin (0) dedicatedsqlpool 00:00:00 Row: 1, Col: 1 5 rows

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